Feature Selection and Model Creation

Feature Selection:

- Selected features based on correlation and SHAP analysis:
    - Top features: MonthlyCharges, tenure, service_count, engagement_score, Contract_Type, tenure_group
- Removed low-importance features

Models Trained:

1. Logistic Regression
2. Random Forest
3. Support Vector Machine (SVM)
4. K-Nearest Neighbors (KNN)
5. Neural Network (MLP)
6. XGBoost
7. LightGBM

Hyperparameter Tuning:

- Optuna used for **XGBoost** optimization
- GridSearchCV or manual tuning for Random Forest, SVM, MLP

Ensemble Learning:

- **Stacking Classifier**: Combined Random Forest + XGBoost + LightGBM with Logistic Regression as final estimator

Evaluation Metrics:

- Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Confusion matrix and ROC curve plotted